Embedding Models for RAG: Comparative Study Report

This task focused on understanding and comparing embedding models used in Retrieval Augmented Generation (RAG) systems. The objective was to move beyond the theory of embeddings and evaluate how different embedding models affect semantic retrieval performance. Three HuggingFace embedding models were downloaded, tested, and compared using a small retrieval dataset.

Methodology

A dataset of 18 technology-related sentences was created. Five semantic queries were designed to test retrieval quality. For each embedding model, the program loaded the model, generated embeddings for the dataset, measured encoding time, recorded vector dimensions, estimated storage requirements, and retrieved the Top-3 most similar sentences for each query using cosine similarity.

Models Evaluated

• all-MiniLM-L6-v2

• all-mpnet-base-v2

• BAAI/bge-small-en-v1.5

Comparison Table

Model	Vector Dim	Avg Encode Time	Storage/1000 Chunks	Top-3 Quality
all-MiniLM-L6-v2	384	0.2541 s	1500 KB	Good
all-mpnet-base-v2	768	1.0313 s	3000 KB	Very Good
BAAI/bge-small-en-v1.5	384	0.3626 s	1500 KB	Excellent

Analysis

The BAAI/bge-small-en-v1.5 model produced the most relevant retrieval results and consistently achieved the highest similarity scores for the test queries. The all-MiniLM-L6-v2 model was the fastest and had low storage requirements, making it efficient for lightweight applications. The all-mpnet-base-v2 model generated larger embeddings and required more computation time, but also provided strong semantic understanding. The results demonstrate a trade-off between retrieval quality and computational cost. Smaller models are faster and require less memory, whereas larger models may improve semantic representation. For resource-constrained RAG systems, all-MiniLM-L6-v2 offers an excellent balance between speed, storage, and accuracy. If maximum retrieval quality is the priority, BAAI/bge-small-en-v1.5 is the preferred choice.

What I Learned

Through this task, I learned how embedding models convert text into dense vector representations that preserve semantic meaning. I gained practical experience with the SentenceTransformers library, generating embeddings, measuring model performance, and implementing semantic retrieval using cosine similarity. I also observed that model selection directly impacts vector size, encoding speed, storage requirements, and retrieval quality. This practical exercise strengthened my understanding of one of the core components of modern RAG systems.

Conclusion

This experiment demonstrated that embedding model selection is an important design decision when building RAG applications. Different models provide different trade-offs between speed, memory usage, and retrieval quality. Based on the experimental results, BAAI/bge-small-en-v1.5 achieved the best overall retrieval performance, while all-MiniLM-L6-v2 was the most efficient option for low-resource environments.